

Supratim Chakraborty

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Education

Jadavpur University

BE in Mechanical Engineering: CGPA: 8.36

Kolkata, India

Sept 2024 - Apr 2028

Experience

Full Stack Developer Intern

Remote

MyndZeal Technologies Pvt. Ltd.

Aug 2025 - Oct 2025

- **Developed 3+ production routes and AI-powered dashboards** using modular components, ensuring seamless UI consistency and responsiveness across all devices.
- **Engineered real-time chat system using WebSockets** with end-to-end encryption to facilitate secure therapist-patient communication. Optimizing backend SQL queries to reduce data latency by 300ms.
- **Integrated Supabase and 3+ external APIs** to automate data ingestion and establish reliable, low-latency data pipelines for core service modules.

Technical Skills

Languages: JavaScript, Python

Frameworks/Libraries: React.js, Node.js, ExpressJS, TailwindCSS, MySQL, Prisma, Postman, NumPy, SciPy, Pandas, Matplotlib, Seaborn, PyTorch, NLTK, HuggingFace,

Tools/Databases: Git, Docker, PostgreSQL, MongoDB, Redis, Vercel, Supabase

Core Competencies: Data Structures and Algorithms, DBMS, CI/CD (Github Actions)

Projects

Agri-Vision Super-Resolution GAN | *PyTorch, ResNet, VGG-19*



April 2026

- Built a **Super-Resolution GAN (SRGAN)** leveraging a 16-block **ResNet** generator and **Sub-Pixel Convolution (PixelShuffle)** to perform 4x upscaling (32x32 to 128x128) of degraded agri-sensor images.
- Optimized a objective function by integrating a frozen **VGG-19 feature extractor for Perceptual Loss**, ensuring high-fidelity biological texture reconstruction alongside L1 Pixel and MSE Adversarial losses.
- Stabilized adversarial training dynamics via a two-phase curriculum, implementing **one-sided label smoothing**, dynamic **instance noise injection**, and **conditional Discriminator updates** to prevent collapse.

Popula | *Gemini API, Scikit-learn, Pandas, Plotly, Streamlit*



Feb 2026 - March 2026

- Built a **hybrid simulation engine** that integrates **Google Gemini** with a **Neural Network** to model policy reactions for up to **50,000 synthetic agents** with unique demographic and economic profiles.
- Engineered a **scalable data pipeline** processing high-dimensional citizen data, featuring real-time **Plotly dashboards** to track social impact metrics like income inequality and population sentiment.
- Developed an **intelligent fallback system** that trains a local model on LLM outputs to handle large-scale batches, reducing **API dependency and latency**.

SciML Molecular Property Predictor | *Transformers, XGBoost, RDKit*



April 2026

- Engineered a multi-modal pipeline using **PyTorch Geometric** (GATv2, GIN, DMPNN) to extract 3D molecular graphs, alongside ChemBERTa and MolFormer for neural NLP embeddings of SMILES/SELFIES.
- Applied **UMAP** and **Truncated SVD** to map high-dimensional sparse/dense features into isolated modalities, training auto-weighted **XGBoost, LightGBM, and CatBoost** specialist ensembles on structural, NLP, and graph compartments.
- Architected a Level-2 contextual meta-learner using Logistic Regression with Test-Time Augmentation (TTA), optimizing the Matthews Correlation Coefficient (MCC) to achieve **0.65 on Private LB**.